



SYSTEM: 7ep

system: "7ep"

services:

- name: "Authentication Service"

responsibilities:

- "Register librarians with username and password"
- "Prevent duplicate librarian registration"
- "Validate credential presence and password strength"
- "Reject short passwords"
- "Authenticate librarian credentials"
- "Validate user sessions"
- "Securely handle credentials without plain-text validation"
- "Provide safe access denial responses"
- "Report authentication operations to audit"

dependencies:

- "Audit Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Book Catalog Service"

responsibilities:

- "Register books in collection by title"
- "Prevent duplicate book registration"
- "Validate title presence during registration"
- "Delete registered books"
- "Prevent deleting unregistered books"
- "Request loan link removal when deleting loaned books"
- "List all registered books"
- "Consult book by identifier"
- "Consult book by title"
- "List books available for loan"
- "Handle empty catalog queries and missing search results"
- "Report catalog operations to audit"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/book-catalog-service"

- name: "Reader Management Service"

responsibilities:

- "Register readers eligible for loans"
- "Prevent duplicate reader registration"
- "Validate reader name during registration"
- "Delete registered readers"
- "Prevent deleting unregistered readers"
- "Request loan link removal when deleting readers with loans"
- "List all registered readers"
- "Consult reader by identifier"
- "Consult reader by name"
- "Reject simultaneous search by identifier and name"
- "Handle empty reader queries and missing search results"
- "Report reader operations to audit"

dependencies:

- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/reader-management-service"

- name: "Loan Management Service"

responsibilities:

- "Register book loans for registered readers"
- "Record loan date"
- "Validate reader existence and current data"
- "Validate book existence and current data"
- "Enforce single-borrower constraint per book"
- "Allow readers to borrow multiple books"
- "Prevent loaning already borrowed books"
- "Remove loan links for deleted books or readers"
- "Preserve consistency among books readers and loans"
- "Report loan registration failures"
- "Report loan operations to audit"

dependencies:

- "Reader Management Service"
- "Book Catalog Service"
- "Audit Service"

suggested_endpoints:

- "/api/loan-management-service"

- name: "Environment Management Service"

responsibilities:

- "Coordinate environment reset across functional areas"
- "Clear data and base structure"
- "Recreate base structure"
- "Restore environment to clean state for automated tests"

dependencies:

- "Authentication Service"
- "Book Catalog Service"
- "Reader Management Service"
- "Loan Management Service"
- "Audit Service"

suggested_endpoints:

- "/api/environment-management-service"

- name: "Demo Utility Service"

responsibilities:

- "Sum two integers"
- "Calculate Fibonacci sequence using multiple approaches"
- "Calculate Ackermann function using multiple approaches"
- "Validate mathematical input formats and report errors"
- "Provide service catalog for endpoint testing"

dependencies:

- ""

suggested_endpoints:

- "/api/demo-utility-service"

- name: "Audit Service"

responsibilities:

- "Provide central event recording mechanism decoupled from functional domains"
- "Record operations from registration authentication consultation and loan flows"
- "Maintain audit trail for traceability problem analysis and failure tracking"

dependencies:

- ""

suggested_endpoints:

- "/api/audit-service"

interactions:

- from: "Authentication Service"
to: "Audit Service"
- from: "Book Catalog Service"
to: "Loan Management Service"
- from: "Book Catalog Service"
to: "Audit Service"
- from: "Reader Management Service"
to: "Loan Management Service"
- from: "Reader Management Service"
to: "Audit Service"
- from: "Loan Management Service"
to: "Reader Management Service"
- from: "Loan Management Service"
to: "Book Catalog Service"
- from: "Loan Management Service"

to: "Audit Service"

- from: "Environment Management Service"

to: "Authentication Service"

- from: "Environment Management Service"

to: "Book Catalog Service"

- from: "Environment Management Service"

to: "Reader Management Service"

- from: "Environment Management Service"

to: "Loan Management Service"

- from: "Environment Management Service"

to: "Audit Service"

SYSTEM: acmeair

system: "acmeair"

services:

- name: "Authentication Service"

responsibilities:

- "Authenticate customers with login and password"
- "Create customer sessions"
- "Validate active sessions and handle session expiration"
- "End sessions and invalidate them on logout"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Flight Search Service"

responsibilities:

- "Manage registered airports"
- "Search one-way and round-trip flights by origin destination and travel date"
- "Provide flight schedules departure and arrival times and reference fares"

dependencies:

- ""

suggested_endpoints:

- "/api/flight-search-service"

- name: "Booking Service"

responsibilities:

- "Create bookings for one-way and round-trip flights"
- "Generate unique booking identifiers"
- "Record booking dates"
- "Consult and list completed bookings"
- "Cancel bookings and update active booking status"

dependencies:

- "Authentication Service"

- "Flight Search Service"

suggested_endpoints:

- "/api/booking-service"

- name: "Customer Profile Service"

responsibilities:

- "Consult customer profile data"
- "Manage customer contact information and address"
- "Update contact and address data with password confirmation"
- "Protect customer personal data against unauthorized access"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-profile-service"

- name: "Loyalty Service"

responsibilities:

- "Provide customer loyalty program status"
- "Present accumulated miles for the current year"
- "Present total accumulated miles as read-only data"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/loyalty-service"

- name: "Environment Administration Service"

responsibilities:

- "Load and reload application database with configurable sample customer data"
- "Recreate operational data including flights customers sessions and test bookings"
- "Provide summarized operational indicators and metrics"
- "Prepare and reset operational data across domain services"

dependencies:

- "Authentication Service"
- "Flight Search Service"
- "Booking Service"
- "Customer Profile Service"

suggested_endpoints:

- "/api/environment-administration-service"

interactions:

- from: "Booking Service"
to: "Authentication Service"
- from: "Booking Service"
to: "Flight Search Service"
- from: "Customer Profile Service"
to: "Authentication Service"
- from: "Loyalty Service"

to: "Authentication Service"

- from: "Environment Administration Service"

to: "Authentication Service"

- from: "Environment Administration Service"

to: "Flight Search Service"

- from: "Environment Administration Service"

to: "Booking Service"

- from: "Environment Administration Service"

to: "Customer Profile Service"

SYSTEM: Cargo-Tracker

system: "Cargo-Tracker"

services:

- name: "Cargo Management Service"

responsibilities:

- "Register new cargo with origin and destination and arrival deadline"
- "Generate unique tracking identifier"
- "Manage cargo lifecycle and routing status"
- "Update cargo destination and trigger route reevaluation"
- "Maintain controlled list of logistics locations"

dependencies:

- "Route Planning Service"

suggested_endpoints:

- "/api/cargo-management-service"

- name: "Route Planning Service"

responsibilities:

- "Retrieve and present candidate routes compatible with cargo specifications"
- "Display route legs including voyages and load and unload points"
- "Assign chosen routes to cargos"
- "Replan routes for off-route cargos or modified destinations"
- "Maintain voyage schedules and route options"

dependencies:

- "Cargo Management Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/route-planning-service"

- name: "Logistics Event Service"

responsibilities:

- "Register logistics events including reception and load and unload and customs and pickup"
- "Validate existence of cargo and location and voyage"
- "Record occurrence and registration timestamps"
- "Isolate rejected registration attempts for operational analysis"
- "Publish verified logistics events for cargo status updates"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Cargo Tracking Service"

suggested_endpoints:

- "/api/logistics-event-service"

- name: "Cargo Tracking Service"

responsibilities:

- "Consolidate cargo operational situation and event history"
- "Track last known location and next expected step and estimated arrival"
- "Evaluate route plan adherence and flag off-route deviations"
- "Provide cargo consultation by tracking identifier"
- "Support geographic tracking visualization highlighting origin and destination and last known location"

dependencies:

- "Cargo Management Service"
- "Route Planning Service"
- "Logistics Event Service"

suggested_endpoints:

- "/api/cargo-tracking-service"

- name: "Integration Service"

responsibilities:

- "Receive logistics events from operational interfaces and integration endpoints and input files"
- "Process periodic file inputs and separate invalid records"
- "Provide consolidated cargo views for monitoring channels"
- "Distribute standardized cargo information to external systems"

dependencies:

- "Cargo Tracking Service"
- "Logistics Event Service"

suggested_endpoints:

- "/api/integration-service"

interactions:

- from: "Cargo Management Service"
to: "Route Planning Service"
- from: "Route Planning Service"
to: "Cargo Management Service"
- from: "Route Planning Service"
to: "Cargo Tracking Service"
- from: "Logistics Event Service"
to: "Cargo Management Service"
- from: "Logistics Event Service"
to: "Route Planning Service"
- from: "Logistics Event Service"
to: "Cargo Tracking Service"

- from: "Cargo Tracking Service"
to: "Cargo Management Service"

- from: "Cargo Tracking Service"
to: "Route Planning Service"

- from: "Cargo Tracking Service"
to: "Logistics Event Service"

- from: "Integration Service"
to: "Cargo Tracking Service"

- from: "Integration Service"
to: "Logistics Event Service"

SYSTEM: DayTrader7

system: "DayTrader7"

services:

- name: "Output must be pure CSV without markdown code blocks"

responsibilities:

- "explanations"

dependencies:

- "or comments.Microservice"

suggested_endpoints:

- "/api/output-must-be-pure-csv-without-markdown-code-blocks"

- name: "Authentication Service"

responsibilities:

- "Authenticate investor credentials"

- "Start and manage active investor sessions"

- "End investor sessions through logout"

- "Validate investor authentication for resource access"

dependencies:

- "Investor Account Service"

suggested_endpoints:

- "/api/authentication-service"

- name: "Investor Account Service"

responsibilities:

- "Register new investors"

- "Validate registration data and unique identification"

- "Maintain investor account balances and access statistics"

- "Consult investor registration and financial data"

- "Update investor profile details"

- "Debit and credit investor balances for trading operations"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/investor-account-service"

- name: "Market Data Service"

responsibilities:

- "Query asset quotes individually and simultaneously"
- "Display asset quote details and variations"
- "Handle invalid asset code queries"
- "Provide consolidated market summary"
- "Update traded volume and market performance indicators"

dependencies:

- "Simulation Administration Service"

suggested_endpoints:

- "/api/market-data-service"

- name: "Trading Service"

responsibilities:

- "Create and process buy orders"
- "Create and process sell orders"
- "Apply fixed operation fees"
- "Manage trading order status lifecycle"
- "Maintain investor order history and recent order views"
- "Alert investors of completed orders"

dependencies:

- "Authentication Service"
- "Investor Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Simulation Administration Service"

suggested_endpoints:

- "/api/trading-service"

- name: "Portfolio Service"

responsibilities:

- "Maintain investor portfolio positions"
- "Register positions upon completed buy orders"
- "Remove positions upon completed sell orders"
- "Consult portfolio positions and details"
- "Calculate invested amounts market values and gains or losses"
- "Provide consolidated financial position summary"
- "Initiate sell orders directly from portfolio positions"

dependencies:

- "Authentication Service"
- "Investor Account Service"
- "Market Data Service"
- "Trading Service"

suggested_endpoints:

- "/api/portfolio-service"

- name: "Simulation Administration Service"

responsibilities:

- "Configure simulation operational parameters"
- "Define order processing mode and simulated capacity limits"
- "Manage market summary refresh intervals and alert settings"
- "Rebuild and reset simulation operational database"
- "Provide summarized execution statistics for users assets orders and positions"

dependencies:

- "Investor Account Service"
- "Market Data Service"
- "Portfolio Service"
- "Trading Service"

suggested_endpoints:

- "/api/simulation-administration-service"

interactions:

- from: "Output must be pure CSV without markdown code blocks"

to: "or comments.Microservice"

- from: "Authentication Service"

to: "Investor Account Service"

- from: "Investor Account Service"

to: "Authentication Service"

- from: "Market Data Service"

to: "Simulation Administration Service"

- from: "Trading Service"

to: "Authentication Service"

- from: "Trading Service"

to: "Investor Account Service"

- from: "Trading Service"

to: "Market Data Service"

- from: "Trading Service"

to: "Portfolio Service"

- from: "Trading Service"

to: "Simulation Administration Service"

- from: "Portfolio Service"

to: "Authentication Service"

- from: "Portfolio Service"

to: "Investor Account Service"

- from: "Portfolio Service"

to: "Market Data Service"

- from: "Portfolio Service"

to: "Trading Service"

- from: "Simulation Administration Service"

to: "Investor Account Service"

- from: "Simulation Administration Service"
to: "Market Data Service"

- from: "Simulation Administration Service"
to: "Portfolio Service"

- from: "Simulation Administration Service"
to: "Trading Service"

SYSTEM: Jokul

system: "Jokul"

services:

- name: "Authentication Service"

responsibilities:

- "Register users with unique usernames"
- "Authenticate user login credentials"
- "Manage user sessions and logout"
- "Differentiate and enforce regular user and administrator profiles"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Catalog Service"

responsibilities:

- "List available movies with pagination support"
- "Provide movie details and full cast"
- "Manage categories and filter movies by category"
- "Register new movies with unique titles"
- "Delete individual and batch movies from the catalog"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Media Service"

responsibilities:

- "Process video file uploads"
- "Manage video file storage and staging from local repository"
- "Serve video content for playback"
- "Provide download links for available movie files"
- "Verify physical video file availability for registered titles"

dependencies:

- "Authentication Service"
- "Catalog Service"

suggested_endpoints:

- "/api/media-service"

interactions:

- from: "Catalog Service"
to: "Authentication Service"
- from: "Media Service"
to: "Authentication Service"
- from: "Media Service"
to: "Catalog Service"

SYSTEM: JpetStore

system: "JpetStore"

services:

- name: "Catalog Service"

responsibilities:

- "Organize products by animal categories"
- "Provide category navigation and product listings"
- "Provide item details and stock status"
- "Provide product keyword search"

dependencies:

- "Inventory Service"

suggested_endpoints:

- "/api/catalog-service"

- name: "Customer Account Service"

responsibilities:

- "Register new customers"
- "Manage profile preferences and favorite categories"
- "Manage customer contact and address data"
- "Provide customer registration information"

dependencies:

- "Authentication Service"

suggested_endpoints:

- "/api/customer-account-service"

- name: "Authentication Service"

responsibilities:

- "Authenticate customers with credentials"
- "Manage login sessions and logout"
- "Provide access control for protected operations"

dependencies:

- ""

suggested_endpoints:

- "/api/authentication-service"

- name: "Shopping Cart Service"

responsibilities:

- "Manage temporary selection of items"

- "Add items and consolidate item quantities"
- "Update and remove cart items"
- "Calculate cart subtotal"
- "Clear customer cart"

dependencies:

- "Catalog Service"
- "Inventory Service"

suggested_endpoints:

- "/api/shopping-cart-service"

- name: "Inventory Service"

responsibilities:

- "Manage item availability"
- "Provide updated stock status and backorder indications"
- "Register stock decrease upon purchase completion"

dependencies:

- "Catalog Service"

suggested_endpoints:

- "/api/inventory-service"

- name: "Order Service"

responsibilities:

- "Manage checkout process and order lifecycle"
- "Compose orders using cart and customer registration data"
- "Manage billing delivery and payment information"
- "Generate unique purchase identifier and record order details"
- "Provide completed order listing and details consultation"

dependencies:

- "Authentication Service"
- "Customer Account Service"
- "Inventory Service"
- "Shopping Cart Service"

suggested_endpoints:

- "/api/order-service"

interactions:

- from: "Catalog Service"
to: "Inventory Service"
- from: "Customer Account Service"
to: "Authentication Service"
- from: "Shopping Cart Service"
to: "Catalog Service"
- from: "Shopping Cart Service"
to: "Inventory Service"
- from: "Inventory Service"
to: "Catalog Service"

- from: "Order Service"
to: "Authentication Service"

- from: "Order Service"
to: "Customer Account Service"

- from: "Order Service"
to: "Inventory Service"

- from: "Order Service"
to: "Shopping Cart Service"

SYSTEM: Pet-Clinic

system: "Pet-Clinic"

services:

- name: "Owner Service"

responsibilities:

- "Register owners"
- "Update owner data"
- "Search owners by last name"
- "List paginated owners"
- "Display owner details"

dependencies:

- ""

suggested_endpoints:

- "/api/owner-service"

- name: "Animal Service"

responsibilities:

- "Register animals linked to owners"
- "Update animal data"
- "Validate animal birth dates and unique names"
- "Provide predefined animal types"

dependencies:

- "Owner Service"

suggested_endpoints:

- "/api/animal-service"

- name: "Visit Service"

responsibilities:

- "Register animal visits"
- "Maintain visit service history"
- "Display visit history of owner animals"
- "Resume pending visit processes"

dependencies:

- "Animal Service"
- "Veterinarian Service"

suggested_endpoints:

- "/api/visit-service"

- name: "Veterinarian Service"

responsibilities:

- "List veterinarians and specialties"
- "Inform next expected visits"
- "Calculate veterinarian workload"
- "Assign visits to veterinarians"

dependencies:

- "Visit Service"

suggested_endpoints:

- "/api/veterinarian-service"

interactions:

- from: "Animal Service"
to: "Owner Service"
- from: "Visit Service"
to: "Animal Service"
- from: "Visit Service"
to: "Veterinarian Service"
- from: "Veterinarian Service"
to: "Visit Service"

SYSTEM: TNTConcept

system: "TNTConcept"

services:

- name: "Administration and Configuration Service"

responsibilities:

- "Maintain support records including departments contract types user categories organization types intera"
- "Maintain individual user configurations"
- "Provide parameter and configuration data to other capabilities"

dependencies:

- ""

suggested_endpoints:

- "/api/administration-and-configuration-service"

- name: "Identity and Access Service"

responsibilities:

- "Register and maintain user personal contractual and department records"
- "Manage passwords administrative resets and forced renewals"
- "Control access by user profile and scope"
- "Provide user and access validation"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/identity-and-access-service"

- name: "Workforce Management Service"

responsibilities:

- "Maintain work agreements and employee records"
- "Manage vacation requests and balance calculations"
- "Track request states and contract expiration"
- "Request notification dispatch for vacation approvals and contract expiration"

dependencies:

- "Identity and Access Service"
- "Administration and Configuration Service"
- "Notification Service"

suggested_endpoints:

- "/api/workforce-management-service"

- name: "Project and Work Tracking Service"

responsibilities:

- "Maintain client-linked projects"
- "Register and classify internal and external activities"
- "Calculate worked hours"
- "Enforce activity evidence policies"

dependencies:

- "Workforce Management Service"
- "Identity and Access Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/project-and-work-tracking-service"

- name: "CRM and Proposals Service"

responsibilities:

- "Maintain organizations contacts and interactions"
- "Manage commercial proposals and proposal composition"
- "Track proposal lifecycle states"
- "Link proposals to projects"

dependencies:

- "Administration and Configuration Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/crm-and-proposals-service"

- name: "Financial Management Service"

responsibilities:

- "Maintain financial accounts and journal entries"
- "Calculate and track financial indicators"
- "Import billable hours and costs"
- "Process payment entries and financial reconciliations"

dependencies:

- "Billing Service"
- "Project and Work Tracking Service"

suggested_endpoints:

- "/api/financial-management-service"

- name: "Billing Service"

responsibilities:

- "Generate and detail invoices"
- "Calculate totals taxes and balances"
- "Convert accepted proposals into billing records"
- "Incorporate billable project hours"
- "Attach invoice documents and billing evidence"

dependencies:

- "CRM and Proposals Service"
- "Project and Work Tracking Service"
- "Financial Management Service"
- "File Storage Service"
- "Administration and Configuration Service"

suggested_endpoints:

- "/api/billing-service"

- name: "Content and Document Management Service"

responsibilities:

- "Organize documents by categories and versions"
- "Associate documents with users and organizational areas"
- "Manage document lifecycles and metadata"

dependencies:

- "File Storage Service"
- "Identity and Access Service"

suggested_endpoints:

- "/api/content-and-document-management-service"

- name: "Commissioning Service"

responsibilities:

- "Register and manage commission orders"
- "Track commission lifecycle and fulfillment workflows"

dependencies:

- "Project and Work Tracking Service"
- "Workforce Management Service"
- "File Storage Service"
- "Notification Service"

suggested_endpoints:

- "/api/commissioning-service"

- name: "Reporting Service"

responsibilities:

- "Consolidate cross-domain business data"

- "Generate and export analytical and operational reports"

dependencies:

- "Identity and Access Service"
- "Workforce Management Service"
- "Project and Work Tracking Service"
- "CRM and Proposals Service"
- "Financial Management Service"
- "Billing Service"
- "Commissioning Service"
- "Content and Document Management Service"

suggested_endpoints:

- "/api/reporting-service"

- name: "Notification Service"

responsibilities:

- "Maintain notification templates and settings"
- "Send email notifications on behalf of business domains"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/notification-service"

- name: "File Storage Service"

responsibilities:

- "Store and retrieve files and binary attachments"
- "Manage storage policies and settings"

dependencies:

- "Administration and Configuration Service"

suggested_endpoints:

- "/api/file-storage-service"

interactions:

- from: "Identity and Access Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Identity and Access Service"
- from: "Workforce Management Service"
to: "Administration and Configuration Service"
- from: "Workforce Management Service"
to: "Notification Service"
- from: "Project and Work Tracking Service"
to: "Workforce Management Service"
- from: "Project and Work Tracking Service"
to: "Identity and Access Service"
- from: "Project and Work Tracking Service"
to: "File Storage Service"

- from: "Project and Work Tracking Service"
to: "Notification Service"

- from: "CRM and Proposals Service"
to: "Administration and Configuration Service"

- from: "CRM and Proposals Service"
to: "Project and Work Tracking Service"

- from: "Financial Management Service"
to: "Billing Service"

- from: "Financial Management Service"
to: "Project and Work Tracking Service"

- from: "Billing Service"
to: "CRM and Proposals Service"

- from: "Billing Service"
to: "Project and Work Tracking Service"

- from: "Billing Service"
to: "Financial Management Service"

- from: "Billing Service"
to: "File Storage Service"

- from: "Billing Service"
to: "Administration and Configuration Service"

- from: "Content and Document Management Service"
to: "File Storage Service"

- from: "Content and Document Management Service"
to: "Identity and Access Service"

- from: "Commissioning Service"
to: "Project and Work Tracking Service"

- from: "Commissioning Service"
to: "Workforce Management Service"

- from: "Commissioning Service"
to: "File Storage Service"

- from: "Commissioning Service"
to: "Notification Service"

- from: "Reporting Service"
to: "Identity and Access Service"

- from: "Reporting Service"
to: "Workforce Management Service"

- from: "Reporting Service"
to: "Project and Work Tracking Service"

- from: "Reporting Service"
to: "CRM and Proposals Service"

- from: "Reporting Service"
to: "Financial Management Service"

- from: "Reporting Service"

to: "Billing Service"

- from: "Reporting Service"

to: "Commissioning Service"

- from: "Reporting Service"

to: "Content and Document Management Service"

- from: "Notification Service"

to: "Administration and Configuration Service"

- from: "File Storage Service"

to: "Administration and Configuration Service"